

SUMMATIVE ASSESSMENT - II

MATHEMATICS

General Instructions:

- i) All questions are compulsory.
- ii) The question paper consists of 31 questions divided into four sections - A, B, C and D.
- iii) Section A contains 4 questions of 1 mark each, Section B contains 6 questions of 2 marks each, Section C contains 10 questions of 3 marks each, and Section D contains 11 questions of 4 marks each.
- iv) Use of calculators is not permitted.

SECTION - A

Question numbers 1 to 4 carry 1 mark each.

1. In Fig.1, PQ is a tangent at point C to a circle with centre O. If AB is a diameter and $\angle CAB = 30^\circ$, find $\angle PCA$.
2. For what value of k will $k + 9$, $2k - 1$ and $2k + 7$ be consecutive terms of an A.P.?
3. A ladder leaning against a wall makes an angle of 60° with the horizontal. If the foot of the ladder is 2.5 m away from the wall, find the length of the ladder.
4. A card is drawn at random from a well-shuffled pack of 52 playing cards. Find the probability of getting neither a red card nor a queen.

SECTION - B

Question numbers 5 to 10 carry 2 marks each.

5. If -5 is a root of the quadratic equation $2x^2 + px - 15 = 0$ and the quadratic equation $p(x^2 + x) + k = 0$ has equal roots, find the value of k .
6. Let P and Q be the points of trisection of the line segment joining A(2,-2) and B(-7,4), such that P is nearer to A. Find the coordinates of P and Q.
7. In Fig.2, a quadrilateral ABCD circumscribes a circle touching sides AB, BC, CD and DA at P, Q, R and S respectively. Prove that:

$$AB + CD = BC + DA$$

8. Prove that the points (3,0), (6,4), and (-1,3) are the vertices of a right-angled isosceles triangle.
9. The 4th term of an A.P. is zero. Prove that the 25th term is three times its 11th term.
10. In Fig.3, from an external point P, two tangents PT and PS are drawn to a circle with centre O and radius r . If $OP = 2r$, show that:

$$\angle OTS = \angle OST = 30^\circ$$

SECTION - C

Question numbers 11 to 20 carry 3 marks each.

11. In Fig.4, O is the centre of a circle such that diameter AB = 13 cm and AC = 12 cm. Find the area of the shaded region. (Take $\pi = 3.14$)
12. In Fig.5, a tent is in the shape of a cylinder with a conical top of same diameter. If the cylindrical part has height 2.1 m and diameter 3 m, and slant height of cone is 2.8 m, find the cost of canvas needed at Rs. 500/sq.m. (Use $\pi = \frac{22}{7}$)
13. If a point P(x,y) is equidistant from the points A(a+b, a-b), prove that:

$$bx = ay$$

14. In Fig.6, find the area of the shaded region enclosed between two concentric circles of radii 7 cm and 14 cm where $\angle AOC = 40^\circ$. (Use $\pi = \frac{22}{7}$)
15. If the ratio of the sum of the first n terms of two A.P.s is $(7n + 1) : (4n + 27)$, find the ratio of their m^{th} terms.
16. Solve:

$$\frac{1}{(x-1)(x-2)} + \frac{1}{(x-2)(x-3)} = \frac{2}{3}, \quad x \neq 1, 2, 3$$

17. A conical vessel with base radius 5 cm and height 24 cm is full of water. This water is emptied into a cylindrical vessel of base radius 10 cm. Find the height to which water rises.
18. A sphere of diameter 12 cm is dropped into a cylindrical vessel partially filled with water. If the water rises by $3\frac{5}{9}$ cm, find the diameter of the vessel.
19. A man standing on the deck of a ship 10 m above water level observes the angle of elevation of the top of a hill as 60° and angle of depression of the base as 30° . Find the distance of the hill from the ship and the height of the hill.
20. Three different coins are tossed together. Find the probability of:
 - (a) exactly two heads
 - (b) at least two heads
 - (c) at least two tails

SECTION - D

Question numbers 21 to 31 carry 4 marks each.

21. Due to heavy floods, 50 schools collectively offered to provide place and canvas for 1500 tents. The lower part of each tent is cylindrical of radius 2.8 m and height 3.5 m, with a conical top of height 2.1 m. If canvas costs Rs.120 per sq.m, find the amount shared by each school. (Use $\pi = \frac{22}{7}$)
22. Prove that the lengths of two tangents drawn from an external point to a circle are equal.
23. Draw a circle of radius 4 cm. Draw two tangents inclined at an angle of 60° to each other.
24. In Fig.7, two equal circles with centres O and O' touch each other. Find:

$$\frac{DO'}{CO}$$

25. Solve:

$$\frac{1}{x+1} + \frac{2}{x+2} + \frac{4}{x+4} = \frac{2}{3}, \quad x \neq -1, -2, -4$$

26. The angle of elevation of the top Q of a vertical tower PQ from point X on the ground is 60° . From a point Y, 40 m vertically above X, the angle of elevation is 45° . Find the height of the tower and distance OX. (Use $\sqrt{3} = 1.73$)
27. The houses in a row are numbered consecutively from 1 to 49. Show that there exists a value of X such that the sum of house numbers preceding X equals the sum of numbers following X.
28. In Fig.8, the vertices of $\triangle ABC$ are A(4,6), B(1,5), and C(7,2). A line segment DE intersects AB and AC such that:

$$\frac{AD}{AB} = \frac{AE}{AC} = \frac{1}{3}$$

29. A number x is selected at random from $\{1,2,3,4\}$ and another number y from $\{1,4,9,16\}$. Find the probability that $xy < 16$.
30. In Fig.9, a sector OAP of a circle with centre O contains angle θ . AB is perpendicular to OA and meets OP produced at B. Prove that the perimeter of the shaded region is:

$$r \left[\tan \theta + \sec \theta + \frac{\pi \theta}{180} - 1 \right]$$

31. A motor boat whose speed is 24 km/h in still water takes 1 hour more to go 32 km upstream than to return downstream. Find the speed of the stream.